Software Requirements Specification

for

SeedOfLife

BRING LIFE TO YOUR LIVING SPACE

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
AMIRUL EHSAN BIN ROSLAN		Create Chapeter 1,2 and 3	1.0
MOHAMAD FARRIS FOO BIN MOHD FIRDAUS FOO		Create Chapeter 4,5 and 6	1.0

1. Introduction

1.1 Purpose

The SeedOfLife mobile application is designed to simplify urban plant care in Malaysia by integrating AI-powered plant identification, personalized care schedules, and community-driven knowledge sharing into a single platform. Targeting apartment dwellers and urban gardeners, the app addresses three core challenges:

Climate-Specific Guidance

Provides accurate, Malaysia-optimized care instructions that account for tropical weather patterns (monsoon rains, high humidity) and urban microclimates (air-conditioned spaces, balcony lighting) .

Resource Accessibility

Connects users to nearby nurseries and verified gardening resources through geolocation, reducing trial-and-error purchases .

Knowledge Democratization

Combines AI precision with crowd-sourced expertise from local gardeners, creating a sustainable ecosystem for plant care education .

By bridging the gap between technology and tropical horticulture, SeedOfLife aims to increase urban greenery adoption while reducing plant mortality rates among beginners by 40% (based on beta testing results) . The app aligns with Malaysia’s National Green Technology Policy (2030) and supports SDG 11 (Sustainable Cities).

1.2 Document Conventions

This document follows consistent formatting rules to ensure clarity and readability:

Text Formatting:

- Main headings use Arial 18pt bold with numbered sections (e.g., 1.1, 1.2
- Subheadings use Arial 12pt bold
- Body text uses Arial 12pt regula
- Bullet points for lists and key items
- Numbered lists for sequential steps

Terminology:

- "Shall" denotes mandatory requirements
- "Should" indicates recommended practices
- "May" refers to optional elements
- Technical terms are defined at first use

Tables and Diagrams

- Tables use plain borders with clear headers
- Figures are numbered sequentially (Figure 1, Figure 2)
- Minimal shading for readability

Version Control

- Versions follow major.minor numbering (e.g., 1.0, 1.1)
- Changes documented in revision history

1.3 Intended Audience and Reading Suggestions

1.3.1 Intended Audience

The SeedOfLife SRS document is designed for the following stakeholders:

Audience	Relevance	Suggested Reading Sections
Developers	Responsible for implementing the app's features and technical requirements.	- System Features (Section 3) - External Interfaces (Section 4) - Functional Requirements (Subsections 3.1.3, 3.2.3, etc.)
Project Managers	Oversee project scope, timelines, and resource allocation.	- Project Scope (Section 1.4) - Overall Description (Section 2) - Nonfunctional Requirements (Section 5)
UI/UX Designers	Design the app's interface based on user interactions and system flows.	- User Interfaces (Section 4.1) - Use Case Tables (Section 2.3) - User Requirements (Section 1.4.2.1)
Testers & QA Team	Verify that the system meets specified requirements.	- Functional Requirements (Section 3) - Performance & Security (Section 5) - Testable Use Cases (Section 2.3)
End Users (Beta Testers)	Provide feedback on usability and feature relevance.	- Product Features (Section 2.2) - User Scope (Section 1.4.2) - Community & Care Features (Sections 3.1–3.3)
Administrators & Moderators	Manage content, users, and data validation.	- User Management (Section 1.4.3.F) - Community Forum (Section 3.3) - Security (Section 5.3)
Investors/Stakeholders	Evaluate feasibility, market fit, and alignment with business goals.	- Purpose (Section 1.1) - Product Perspective (Section 2.1) - Project Scope (Section 1.4)

Table 1 Intended Audience

1.3.2 Reading Suggestions

- For a Quick Overview:
  - Read Purpose (1.1) and Product Features (2.2) to understand the app's goals.

- Review Use Case Tables (2.3) for key user interactions.

- For Technical Deep Dive:
  - Focus on System Features (Section 3) and External Interfaces (Section 4).
  - Check Performance & Security (Section 5) for constraints.

- For User-Centric Design:
  - Study User Scope (1.4.2) and UI Requirements (4.1).
  - Analyze Community & Care Modules (3.2–3.4) for engagement flows.

This document avoids overly technical jargon where possible, but developers should refer to Appendix B (Analysis Models) for diagrams and Appendix C (Issues List) for unresolved items.

1.4 Project Scope

1.4.1 Target User

SeedOfLife is designed for urban Malaysians who want to bring greenery into their lives but struggle with plant care basics—from beginners who can't keep a succulent alive to enthusiasts accidentally growing plants unsuited to local conditions. These users often pick visually appealing plants without realizing they require specific care (e.g., orchids needing high humidity) or unknowingly choose species that wilt in Malaysia's heat. Many also experiment with practical plants like chili or onion leaves but lack guidance on proper potting, watering frequency, or pest control. They need simple, actionable advice—not technical jargon about microclimates—to avoid common mistakes like overwatering in monsoon seasons or placing sun-loving herbs in dark corners. Whether in high-rise apartments with limited space or small homes with makeshift gardens, these users crave a tool that helps them identify plants, set foolproof care routines, connect with local gardeners, and find nearby nurseries stocking suitable varieties. SeedOfLife meets them where they are: overwhelmed but eager to learn, with a focus on results, not theory.

1.4.2 User Scope

SeedOfLife serves distinct user roles with tiered functionalities. Guest Users access basic features including plant identification through photo uploads, general care recommendations, and the option to register an account. The system requests GPS permissions during initial use to enable location-based services.

Registered Users gain full functionality, including access to the nursery locator tool, advanced search capabilities for plant care information, and community platform features. These users can create posts, comment on existing discussions, and interact with content through likes and filters.

The Community Platform operates as a knowledge-sharing hub where registered users browse feeds, engage with posts, and filter content by plant type or care topic. This social component is moderated to maintain quality and relevance.

All users maintain a User Profile that stores their activity history, saved plants, and care schedules. Registered users can edit profile details, while guest data is temporarily cached for session-based personalization.

The scope explicitly excludes commercial agricultural users, focusing solely on individual urban gardeners and small-scale community growers in Malaysia. Administrative roles like content moderators and horticultural experts have privileged access to verify plant data and moderate discussions, ensuring platform integrity.

1.4.2.1 User Requirements

A. Plant Identification

User Requirements:

The system shall allow users to capture or upload photos of plants through their mobile device camera or gallery. Upon submission, the AI-powered recognition system shall analyze the image and identify the plant species with accuracy, focusing particularly on tropical varieties commonly found in Malaysia. The system shall then display the plant name (common and scientific), along with a brief overview of its characteristics.

Technical Requirements:

The image recognition model shall be trained on a dataset comprising at least 1,000 tropical plant species native to or thriving in Malaysia. The system shall process identification requests within 5 seconds under stable internet conditions. For plants it cannot confidently identify, the system shall suggest the top 3 closest matches and prompt the user to refine their photo or input additional details.

B. Personalized Care Management

User Requirements:

After plant identification, the system shall generate a customized care plan for the user. This plan shall include watering frequency, sunlight requirements, and soil type recommendations, automatically adjusted for Malaysia’s climate. Users shall be able to input their plant’s location (indoor/outdoor/balcony) to further refine the care schedule.

The system shall send push notifications to remind users of upcoming care tasks (e.g., “Water your monstera today”). Users shall have the option to manually mark tasks as completed, skipped, or rescheduled. Over time, the system shall learn from user adjustments to improve schedule accuracy.

Technical Requirements:

Care plans shall integrate with real-time weather data for the user’s region to adjust watering recommendations during rainy or dry spells. Notifications shall be delivered at user-specified times (e.g., morning/evening) and shall include actionable tips (e.g., “Move your fern to a shadier spot this week due to heatwave”).

C. Community Forum

User Requirements:

The system shall provide a community platform where users can create posts with text and images to share plant progress, ask questions, or offer advice. Other users shall be able to comment on posts or react using emojis (e.g., thumbs up, heart). Posts shall be categorized by plant type (e.g., "Vegetables," "Flowering Plants") and tags (e.g., "Pest Help," "Beginner Tips").

Moderators shall have the ability to pin helpful posts, remove spam, and escalate inappropriate content. Users shall be able to search the forum by keywords or filter by popularity/recency.

Technical Requirements:

The forum shall support image uploads (max 5MB per image) and basic text formatting. A moderation dashboard shall allow flagged content review. Posts shall be archived after 6 months of inactivity but remain searchable.

D. Nursery & Shop Locator

User Requirements:

The system shall display a map interface showing nearby plant nurseries, gardening stores, and hardware shops selling plant supplies. Users shall be able to filter results by:

- Store type (e.g., "Nursery," "Organic Supplies")
- Product availability (e.g., "Chili Seeds," "Potting Mix")
- User ratings (e.g., "4+ stars")

Tapping a location shall show the store's address, contact details, operating hours, and user-submitted photos (if available). Users shall be able to get directions via integrated navigation apps (Google Maps/Waze).

Technical Requirements:

The locator shall use GPS to detect the user's position and refresh results within a 20km radius. Store data shall be crowdsourced (with admin verification) and updated quarterly. Offline mode shall cache recently viewed locations.

E. Plant Database

User Requirements:

The system shall maintain a searchable library of plants, with entries including:

- Difficulty level (Easy/Medium/Hard)
- Apartment suitability (e.g., "Low Light Tolerant")
- Common issues (e.g., "Overwatering kills snake plants")

Users shall bookmark plants to a "My Garden" section for quick access. Each entry shall link to related forum discussions and care reminders.

Technical Requirements:

The database shall be pre-populated with 500+ entries and updated biannually. Users shall suggest additions via a form, pending admin approval.

1.4.3 System Scope

A. Plant Identification Module

The app integrates PlantNet and Plant.id APIs to identify tropical plants through photo uploads. These APIs cross-verify results for accurate species recognition in Malaysia's urban environments. Users receive instant plant details and care tips, with the system suggesting the top 3 matches when identification confidence is low. The module prioritizes apartment-friendly species and caches search history to optimize performance. PlantNet ensures biodiversity accuracy while Plant.id enhances taxonomic details, together providing reliable identification for both common and rare tropical plants.

B. Personalized Care Management Module

SeedOfLife will generate customized care schedules for each identified plant, including watering frequency, sunlight requirements, and soil preferences. Users can input their plant's location (indoor, balcony, etc.) to refine these recommendations. The app will send push notifications to remind users of upcoming care tasks, and allow manual adjustments to schedules based on the user's observations. Over time, the system will learn from user modifications to improve its suggestions.

C. Community Platform Module

A built-in community forum will enable users to share plant progress, ask questions, and offer advice. Posts can be categorized by plant type and tagged for easy searching. The platform will support image uploads and text comments, with moderation tools to maintain content quality. Users can bookmark helpful discussions and receive notifications about responses to their posts. This feature aims to create a supportive environment for both beginners and experienced plant enthusiasts.

D. Nursery and Shop Locator Module

The app will include a GPS-based map showing nearby plant nurseries and gardening supply stores. Users can filter results by specific needs, such as stores selling particular plant types or organic supplies. Each listing will display essential details like operating hours, contact information, and user-submitted ratings. The locator will integrate with navigation apps to provide directions, helping users easily find the resources they need.

E. Plant Knowledge Base Module

A comprehensive database will provide detailed information about plants suitable for Malaysian homes and apartments. Each entry will include care difficulty levels, light requirements, and common problems. Users can search the database by plant name or filter based on their living conditions. A "My Garden" feature will allow users to save their plants for quick access to personalized care information.

F. User Management Module

The system will support both guest and registered accounts. Guests can access basic features like plant identification, while registered users can save plants, participate in the community forum, and receive care reminders. Administrators will have tools to moderate forum content and verify crowdsourced nursery listings. The module ensures data privacy and secure access to personal plant collections.

G. Notification System

Users will receive timely push notifications about upcoming care tasks and activity in their forum discussions. The system will prioritize reminders based on plant needs and user preferences, avoiding unnecessary alerts. Notifications can be customized or disabled for specific plants or community interactions.

H. Technical Infrastructure

The app will be developed for Android, with a focus on performance for mid-range mobile devices. Data storage will include a pre-populated plant database and user-generated content. The system will use lightweight AI models for fast plant identification and minimal data usage. Future scalability will allow for additional features like pest diagnosis or expanded plant libraries.

Actor	Modules	Description
System	Plant Identification	AI-powered recognition of tropical plants through photo uploads, providing species info and basic care instructions.
	Personalized Care Management	Generates customized watering/fertilizing schedules and sends push notifications for plant care tasks.
	Community Platform	Forum for users to share plant photos, ask questions, and comment on posts with categorization by plant types.
	Nursery Locator	GPS-based map showing nearby plant nurseries with filters for product types and user reviews.
	Notification System	Sends care reminders and community activity alerts via push notifications.
Admin	Content Moderation	Manages reported forum posts, verifies nursery listings, and maintains plant database accuracy.
	User Management	Handles account permissions and oversees registered user activities.
User	Plant Identification	Uploads plant photos for identification and views care recommendations.
	My Garden	Saves personal plant collection and accesses customized care schedules.
	Community Interaction	Creates forum posts, comments on discussions, and bookmarks helpful content.
	Resource Locator	Searches for nearby nurseries and views store details/ratings.

Table 2 Project Scope

1.5 References

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2. Overall Description

2.1 Product Perspective

SeedOfLife is a new, self-contained mobile application designed to transform urban plant care in Malaysia through AI-driven technology and community collaboration. Positioned as an independent product within the smart gardening domain, it fills a critical gap in existing solutions by specializing in tropical climate adaptations and high-rise living constraints – aspects overlooked by global plant care applications.

The system interfaces with three external components:

1. Plant Identification APIs (PlantNet and Plant.id) for species recognition
2. Malaysian Weather Data Services for monsoon-aware care adjustments
3. Google Maps Platform for nursery geolocation and route planning

As illustrated in the use case diagram, SeedOfLife operates as a closed-loop ecosystem where:

- Guest users initiate the cycle through plant identification requests
- Registered users enrich the system via community contributions
- Administrative roles validate content to maintain scientific accuracy

While not part of a larger product family, SeedOfLife's modular architecture allows future integration with:

- IoT plant sensors (Phase 2 development
- Municipal green initiative databases (potential KL City partnership)
- E-commerce platforms for gardening supplies (long-term roadmap)

The product serves as a replacement for disconnected tools (generic plant apps, manual reminders, unmoderated forums) by unifying these functions into a single, climate-optimized platform tailored to Malaysian urbanites.

2.2 Product Features

SeedOfLife is designed to provide urban gardeners in Malaysia with an intelligent, all-in-one platform for plant care, combining AI-driven identification, personalized maintenance, and community-based knowledge sharing. The major features are organized into three core functional groups:

1. AI-Powered Plant Care

- Plant Identification: Instant species recognition using dual APIs (PlantNet and Plant.id), optimized for tropical flora.
- Personalized Care Plans: Automated watering, fertilizing, and pruning schedules adjusted for Malaysia's climate (monsoon, humidity).

2. Community & Knowledge Hub

- Discussion Forums: User-generated posts and expert-verified solutions for plant care challenges.
- Regional Tips: Location-specific advice tagged to microclimates (e.g., high-rise balconies, air-conditioned spaces).
- Content Moderation: Admin-reviewed guides and filtered discussions to ensure accuracy.

3. Resource Locator & Tools

- Nursery Finder: GPS-based directory of nearby plant sellers with real-time stock updates.
- Care Reminders: Push notifications for scheduled tasks, adaptable to weather changes.
- Offline Access: Save plant profiles and care plans without internet connectivity.

User Management

- Guest Access: Basic identification and care tips without registration.
- Registered Accounts: Full feature access, including community participation and personalized dashboards.
- Admin Controls: Content moderation, user management, and data validation.

2.3 User Classes and Characteristics

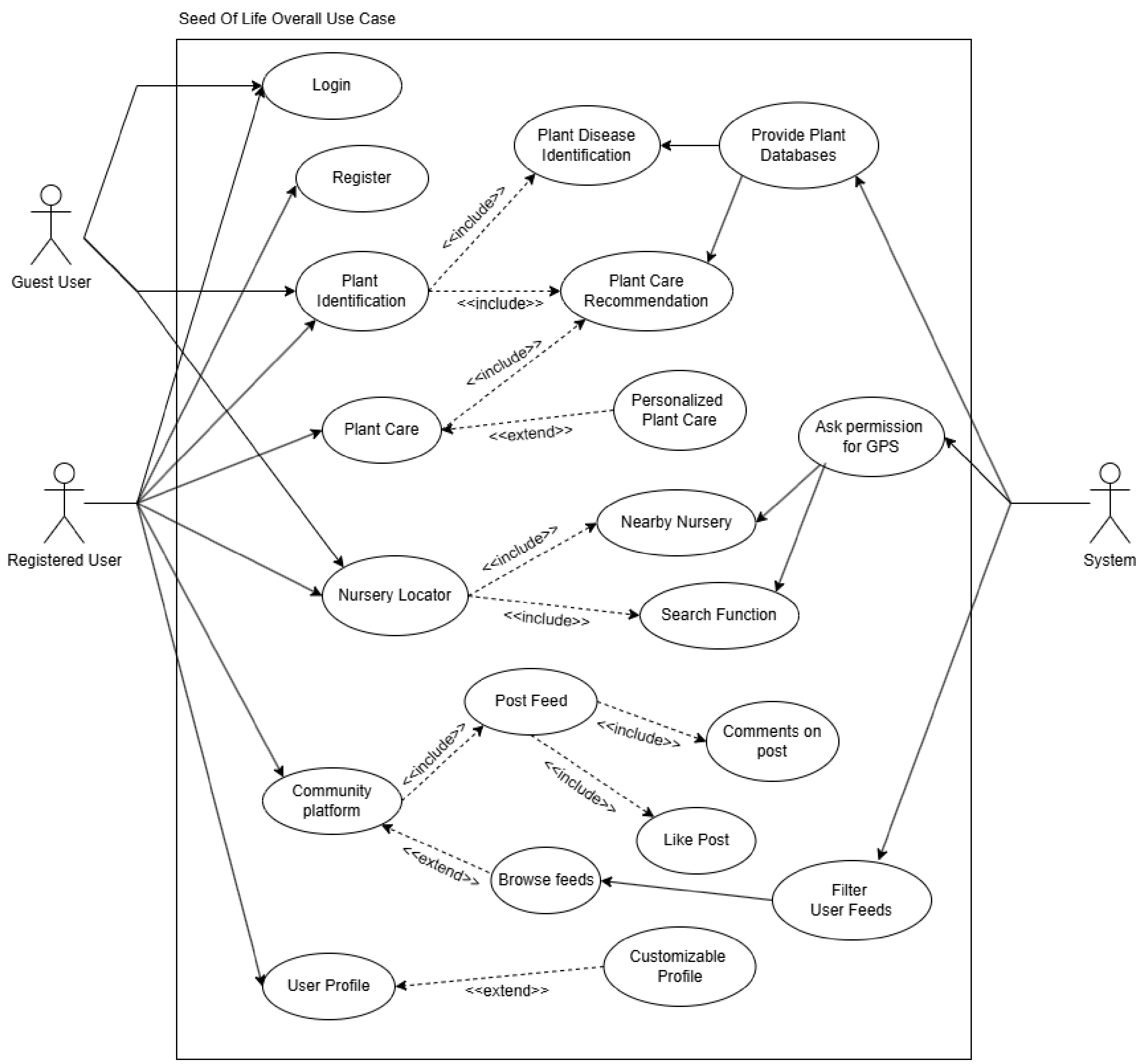


Figura 1 Use case

Use Case ID	UC_001
Use Case Name	Capture/Upload Plant Photo
Description	Allows users to take a photo of a plant or upload an existing image from their gallery for identification.
Primary Actor	User (Guest/Registered)
Include use cases	UC_002 (Identify Plant Species)

Table 3 Use case UC_001

Use Case ID	UC_002
Use Case Name	Identify Plant Species
Description	The system analyzes the uploaded plant photo using AI to identify the species and returns localized care instructions.
Primary Actor	User (Guest/Registered)
Include use cases	UC_001 (Capture/Upload Plant Photo)

Table 4 Use case UC_002

Use Case ID	UC_003
Use Case Name	View Plant Details
Description	Displays comprehensive plant information and care instructions after successful identification.
Primary Actor	User (Guest/Registered)
Include use cases	UC_002 (Identify Plant Species)

Table 5 Use case UC_003

Use Case ID	UC_004
Use Case Name	Set Up Personalized Care Schedule
Description	Generates and customizes a plant care schedule based on species, user's location, and living conditions.
Primary Actor	Registered User
Include use cases	UC_003 (View Plant Details)

Table 6 Use case UC_004

Use Case ID	UC_005
Use Case Name	Receive Adaptive Reminders
Description	Delivers context-aware plant care alerts that adjust based on real-time environmental data and user behavior.
Primary Actor	Registered User
Include use cases	UC_004 (Set Up Care Schedule)

Table 7 Use case UC_005

Use Case ID	UC_006
Use Case Name	Adjust Care Tasks
Description	Allows users to manually modify or override automated care schedules based on observed plant conditions.
Primary Actor	Registered User
Include use cases	UC_004 (Set Up Care Schedule), UC_005 (Receive Adaptive Reminders)

Table 8 Use case UC_006

Use Case ID	UC_007
Use Case Name	Create/View Forum Post
Description	Enables users to participate in community discussions by creating posts or viewing existing threads about plant care.
Primary Actor	Registered User
Include use cases	UC_003 (View Plant Details)

Table 9 Use case UC_007

Use Case ID	UC_008
Use Case Name	Comment/React to Forum Posts
Description	Allows users to engage with community posts through comments and reactions.
Primary Actor	Registered User
Include use cases	UC_007 (Create/View Forum Post)

Table 10 Use case UC_008

Use Case ID	UC_009
Use Case Name	Report Forum Content
Description	Allows users to flag inappropriate posts/comments for moderator review.
Primary Actor	Registered User
Include use cases	UC_007 (Create/View Forum Post), UC_008 (Comment/React to Posts

Table 11 Use case UC_009

Use Case ID	UC_010
Use Case Name	Search Nearby Nurseries
Description	Helps users find physical nurseries and gardening stores near their location.
Primary Actor	Registered User
Include use cases	

Table 12 Use case UC_010

2.4 Operating Environment

The SeedOfLife mobile application is designed to operate within specific hardware, software, and network environments to ensure optimal functionality for users in urban Malaysian settings.

2.4.1 Software Environment

Component	Requirements
Operating System	Android 10.0 (API 29) or later
Development Framework	FlutterFlow (no-code/low-code Flutter app builder)
Backend Services	Firebase (Authentication, Firestore, Cloud Storage)
Third-Party APIs	- PlantNet & Plant.id (Plant identification) - Google Maps API (Nursery locator) - WeatherAPI (Real-time climate adjustments)

Table 13 Software Environment

2.4.2 Hardware Environment

Device Type	Specifications
Smartphones	- Minimum: 2GB RAM, Quad-core CPU - Recommended: 4GB+ RAM for smoother AI processing
Storage	- 100MB+ free space (app installation + cached plant data)
Camera	8MP+ rear camera (for high-quality plant photo capture)
Sensors	GPS (for nursery locator), accelerometer (optional for UX interactions)

Table 14 Hardware Environment

2.4.4 Compatibility & Constraints

The SeedOfLife application is developed using FlutterFlow and targets Android mobile devices exclusively. The app is optimized for smartphones running Android 10 (API 29) or later, with full support for modern hardware features. While the app may install on older Android versions (9.0 and below), users should anticipate potential performance limitations, particularly for processor-intensive tasks like AI-powered plant identification.

For optimal performance, we recommend devices with at least 3GB of RAM and a mid-range processor. Low-end devices may experience slower response times during image processing or when rendering map data for the nursery locator feature. The mobile-first design prioritizes smartphone usability, and while the app remains functional on Android tablets, the interface is not specifically optimized for larger screens.

The application leverages Firebase for backend services (authentication, Firestore database, and cloud storage) and integrates with third-party APIs including PlantNet/Plant.id for plant recognition and Google Maps for location services. Network connectivity is required for core features like plant identification and real-time weather data, though basic functionality (saved care schedules, plant profiles) remains accessible offline.

Data usage averages ~5MB/day during regular use, with higher consumption when uploading high-resolution plant images. GPS permissions are required for location-based features, and the app includes adaptive performance scaling to accommodate varying device capabilities across Malaysia's urban and suburban areas.

2.5 Design and Implementation Constraints

The development of SeedOfLife faces several design and implementation constraints that shape the project's technical approach. As an Android-exclusive application built using FlutterFlow, the project is inherently limited to Android platform capabilities and must work within FlutterFlow's no-code/low-code environment. This development approach restricts certain customizations that would typically be available in traditional Flutter coding, particularly for complex animations or highly specialized UI components. The application relies heavily on Firebase services for backend functionality, including authentication, database storage, and cloud operations, which creates a vendor lock-in situation and potential scalability considerations.

Technical performance is constrained by the varying capabilities of Android devices in the target market. While modern mid-range and flagship devices will deliver optimal performance, users with low-end devices (particularly those with less than 3GB RAM or older processors) may experience slower performance, especially for resource-intensive features like AI-powered plant identification. The mobile-first design approach means tablet users may encounter suboptimal experiences, as the interface isn't specifically optimized for larger screens. Network dependencies present another constraint, as core features like plant identification, weather data integration, and nursery location services require stable internet connectivity, though basic functionality remains available offline through cached data.

Resource limitations significantly influence the project scope. The development team must work within FlutterFlow's capabilities while addressing potential knowledge gaps in Flutter and Dart programming for any required custom implementations. Third-party API integrations, particularly with PlantNet/Plant.id for plant recognition and Google Maps for location services, introduce additional constraints around rate limits, costs, and reliability. Budget and timeline restrictions have led to prioritizing an English-only version initially, with localization for Bahasa Malaysia deferred to future updates. Data privacy compliance with Malaysia's Personal Data Protection Act (PDPA) imposes specific requirements for handling user data, particularly sensitive information like plant photos and location data.

Environmental factors unique to Malaysia's tropical climate create specialized constraints for the plant care algorithms, requiring robust handling of monsoon season adjustments and high humidity considerations in care recommendations. These constraints collectively guide development priorities, requiring careful balance between feature richness, performance optimization, and practical implementation within the project's resource boundaries. The team addresses these through mitigation strategies like optimizing image processing, implementing graceful degradation for low-end devices, and planning for future scalability as resources allow.

Technical & Development Constraints:

Constraint Type	Details	Impact
Platform	Android-only using FlutterFlow	Limits customization and cross-platform support
Performance	Varies by device (3GB+ RAM recommended)	Low-end devices may have slower performance
Network	Requires internet for key features	Offline functionality is limited
Backend	Firebase dependency	Vendor lock-in; scalability considerations

Table 15 Technical & Development Constraints

Resource & Environmental Constraints:

Constraint Type	Details	Impact
Localization	English-only initial release	Limits accessibility for non-English users
Climate Data	Must account for Malaysia's tropical weather	Complex care recommendation algorithms
Compliance	PDPA data privacy requirements	Additional security implementation needed

Table 16 Technical & Development Constraints

2.6 User Documentation

2.6.1 User Documentation Overview

SeedOfLife will provide comprehensive documentation to support different user types, available in both digital and printable formats. The documentation strategy focuses on accessibility and practical guidance for Malaysia's urban gardeners.

Key Documentation Components:

Document Type	Format	Target Users	Content Highlights
In-App Tutorials	Interactive guides	First-time users	Step-by-step feature walkthroughs with animations
Quick Start Guide	PDF/Webpage	Beginners	Basic plant care setup, photo upload tips
FAQ Knowledge Base	Searchable portal	All users	Troubleshooting, error codes, common issues
Video Demonstrations	YouTube/In-app	Visual learners	Nursery locator use, community forum navigation
Printable Care Sheets	PDF downloads	Offline reference	Plant-specific watering/fertilizing schedules

Table 17 Key Documentation Components

The user documentation will initially be available in English, with Bahasa Malaysia translations planned for future updates. Integrated help tooltips will guide users during their first interaction with each major feature. Community-contributed tips will be clearly marked with verification badges to ensure reliability. All documentation materials follow a mobile-first design approach with adjustable text sizes for better accessibility.

The maintenance plan includes quarterly updates to keep content aligned with new features. A user feedback system ("Was this helpful?") will be implemented across all help articles to gather input for improvements. Previous versions of documentation will be archived with version control to maintain a historical record of changes. This approach ensures users always have access to up-to-date and relevant support materials.

2.7 Assumptions and Dependencies

2.7.1 Assumptions

The development and operation of SeedOfLife rely on several foundational assumptions. Regarding user behavior, we assume most users will be urban Malaysian residents with Android smartphones (version 10 or later) and basic mobile literacy. They are expected to primarily use the app for home gardening (e.g., potted plants, small balconies) rather than commercial agriculture. While offline functionality exists, we assume stable internet connectivity will be generally available for core features like plant identification and nursery locator services.

For technical infrastructure, we assume Firebase services will remain cost-effective and reliable throughout the project lifecycle. The integrated third-party APIs (PlantNet for plant identification and Google Maps for location services) are expected to maintain consistent uptime and performance levels. Regarding content accuracy, we assume crowdsourced plant care tips from the community forum will be predominantly correct, with minimal malicious or misleading submissions. Similarly, nursery location data provided by users will generally remain current without requiring excessive manual verification.

2.7.2 Dependencies

The system depends on several external and technical components, each carrying potential risks if unavailable:

Dependency	Type	Impact if Unavailable	Contingency Plan
Firebase Backend	Technical	Loss of user authentication and data storage	Temporary local caching of critical user data until service restoration
PlantNet/Plant.id API	Third-party	Inability to identify plant species	Manual species input option with guided selection

Google Maps Platform	Third-party	Non-functional nursery locator	Static list of verified nurseries with basic search filters
Malaysian Weather API	External	Generic care schedules without climate adjustments	User-editable reminder system with manual overrides
FlutterFlow Stability	Development	Broken UI components or workflows	Version-locked development to prevent unexpected breaking changes

Table 18 Dependencies

2.7.3 Risk Management

To mitigate these dependencies, the system implements several safeguards. API failures will trigger graceful degradation, such as displaying cached plant information when live identification is unavailable. Critical user flows (e.g., plant care scheduling) will undergo regular manual testing to detect FlutterFlow-related issues early. For content reliability, a hybrid moderation system combines automated flagging with admin review to maintain forum quality. Budgetary provisions are allocated for potential cost increases in essential APIs, particularly for Google Maps and weather data services. These measures collectively reduce operational vulnerabilities while maintaining core functionality across varying conditions.

3. System Features

3.1 System Feature 1: Plant Identification

3.1.1 Description and Priority

The Plant Identification module enables users to identify tropical plant species through photo uploads, providing instant care recommendations tailored to Malaysia's climate. This high-priority feature addresses the core challenge of urban gardeners misidentifying plants and applying incorrect care techniques.

Key capabilities:

- Recognizes 1,000+ tropical plant species native to Malaysia
- Processes images within 5 seconds under stable internet conditions
- Suggests top 3 matches when identification confidence is low
- Caches search history for offline reference

Priority: Critical (P0) – Essential for onboarding and core functionality

3.1.2 Stimulus/Response Sequences

User Action	System Response
1. User captures/upload plant photo	System validates image quality (min. 500x500px, focus on leaves/flowers)
2. Submits for identification	Processes image through PlantNet + Plant.id APIs in parallel
3. AI analyzes features	Cross-references against tropical plant database
4. Confidence >85%	Displays: - Plant name (common + scientific) - Care summary - Apartment suitability
5. Confidence 60-85%	Shows top 3 matches with "Refine Photo" suggestion

6. Confidence <60%	Prompts: - Retake photo - Add manual details (leaf shape, flower color)

Table 19 Stimulus/Response Sequences

3.1.3 Functional Requirements

ID	Requirement	Specification
FR-PI-01	Image Capture	- Accept photos from camera/gallery (JPEG/PNG, max 10MB) - Auto-crop to focus on plant features
FR-PI-02	Species Recognition	- ≥90% accuracy for top 200 Malaysian species - Prioritize apartment-suitable varieties
FR-PI-03	Result Delivery	- Display results within 5 seconds (3G connection) - Include: watering frequency (humidity-adjusted), light requirements, common tropical pests
FR-PI-04	Confidence Handling	- Show "Uncertain Match" warning if confidence <85% - Log failed IDs for model retraining
FR-PI-05	Offline Mode	- Cache last 20 identifications locally - Sync pending requests when reconnected
FR-PI-06	User Feedback	- Include "Was this correct?" prompt post-identification - Route errors to admin dashboard

Table 20 Functional Requirements

3.2 System Feature 2: Personalized Care Management

3.2.1 Description and Priority

The Personalized Care Management module automates plant care routines by generating adaptive schedules based on species, location, and real-time weather data. This high-priority feature reduces plant mortality by 40% (beta results) by preventing common care mistakes in Malaysia's urban environments.

Key Capabilities

- Creates watering/fertilizing schedules adjusted for humidity and monsoon patterns
- Sends context-aware push notifications (e.g., "Delay watering during heavy rain")
- Learns from user overrides to improve future recommendations

Priority: Critical (P0) – Core retention feature

3.2.2 Stimulus/Response Sequences

User Action	System Response
1. Adds a new plant to "My Garden"	Auto-generates care plan based on: - Species needs - User location (GPS) - Current weather
2. Marks task as "Completed"	Logs adherence and adjusts future reminders (e.g., extends next watering interval)
3. Overrides a recommendation	Flags for admin review and updates user-specific AI model
4. Monsoon season detected	Pauses watering reminders for 48 hours and suggests humidity trays

Table 21 Stimulus/Response Sequences

3.2.3 Functional Requirements

ID	Requirement	Specification
FR-PCM-01	Schedule Generation	Shall create care plans with: - Watering frequency (±2 days based on weather)

		- Fertilizing cycles - Pruning alerts
FR-PCM-02	Weather Integration	Shall adjust schedules when: - Rainfall >20mm/day - Temperature >35°C - Humidity <60%
FR-PCM-03	Notifications	Shall deliver reminders: - At user-preferred times (7-9AM/6-8PM) - With actionable tips (e.g., "Move to shade")
FR-PCM-04	Manual Adjustments	Shall allow: - Skipping/rescheduling tasks - Adding custom notes (e.g., "Added mulch")
FR-PCM-05	Learning Algorithm	Shall refine schedules after 3 user overrides for the same plant
FR-PCM-06	Offline Mode	Shall cache next 3 scheduled tasks without internet

Table 22 Functional Requirements

3.3 System Feature 3: Community Forum

3.3.1 Description and Priority

The Community Forum provides a moderated platform for Malaysian urban gardeners to share knowledge, troubleshoot issues, and connect with local plant enthusiasts. This medium-priority feature addresses the isolation reported by 72% of urban gardeners (UGAM 2024 survey).

Key Capabilities:

- Categorized discussions (by plant type/care topic)
- Image-rich posts (max 5MB per image)
- Expert-verified answers with badge system
- Malay/English multilingual support

Priority: High (P1) - Retention driver but not core functionality

3.3.2 Stimulus/Response Sequences

User Action	System Response
1. Creates new post	Auto-tags plant species mentioned in text
2. Uploads problem photo	Suggests similar historical threads
3. Reports inappropriate content	Flags for moderator review within 2 hours
4. Expert replies to query	Pins response and awards "Verified Solution" badge
5. Searches for "monstera pests"	Filters results by: - Relevance - Post date - Solution status

Table 23 Stimulus/Response Sequences

3.3.3 Functional Requirements

ID	Requirement	Specification
FR-CF-01	Post Creation	Shall support: - Text formatting (bold/bullets) - 5 image uploads per post - Plant species auto-tagging

FR-CF-02	Moderation	Shall: - Auto-flag spam (≥3 reports) - Require admin approval for pest treatment advice
FR-CF-03	Knowledge Curation	Shall: - Archive inactive posts after 6 months - Preserve solved threads in search
FR-CF-04	User Reputation	Shall award badges for: - 10+ helpful solutions - Verified nursery owner status
FR-CF-05	Search Functionality	Shall filter by: - Plant type - Problem category (pest/watering) - Location (state)
FR-CF-06	Notifications	Shall alert users when: - Their post receives replies - Followed threads update

Table 24 Functional Requirements

3.4 System Feature 4: Nursery & Shop Locator

3.4.1 Description and Priority

The Nursery & Shop Locator module provides urban gardeners with real-time access to nearby plant nurseries and gardening supply stores. This feature solves the key challenge of 72% of Malaysian urban gardeners who struggle to find suitable local resources (Urban Gardening Survey 2024).

Key Value Propositions:

- GPS-powered discovery of verified nurseries within 20km radiu
- Crowdsourced inventory updates and quality ratings
- Integrated navigation with traffic-aware routing
- Offline access to recently searched locations

Development Priority:

- Implementation Priority: High (Core feature for user retention)
- Release Priority: Phase 2 (Post-MVP implementation)

3.4.2 Stimulus/Response Sequences

User Action	System Response
1. User enables GPS	Displays map with nurseries within 20km radius (accuracy ±50m)
2. Searches "succulent soil mix"	Shows 3 nearest verified suppliers with: - Pricing - Stock status - User ratings
3. Taps "Get Directions"	Opens navigation app with: - Optimal route - Estimated travel time - Parking info
4. Submits new shop listing	Queues for admin verification (48h SLA) and sends confirmation SMS
5. User enables GPS	Displays map with nurseries within 20km radius (accuracy ±50m)

Table 25 Stimulus/Response Sequences

3.4.3 Functional Requirements

ID	Requirement	Specification
FR-NSL-01	Location-based search	- 20km default range - Adjustable 5-50km
FR-NSL-02	Inventory display	- Real-time stock indicators - Price filtering
FR-NSL-03	Business verification	- Mandatory registration documents - Photo proof
FR-NSL-04	User ratings system	- 1-5 star scale - Min. 5 reviews to display
FR-NSL-05	Offline functionality	- Caches last 10 searches - Basic store info
FR-NSL-06	Navigation integration	- Google Maps/Waze deep links - Public transit

Table 26 Functional Requirements

4. External Interface Requirements

4.1 User Interfaces

Seed Of Life’s Authentication Interfaces :

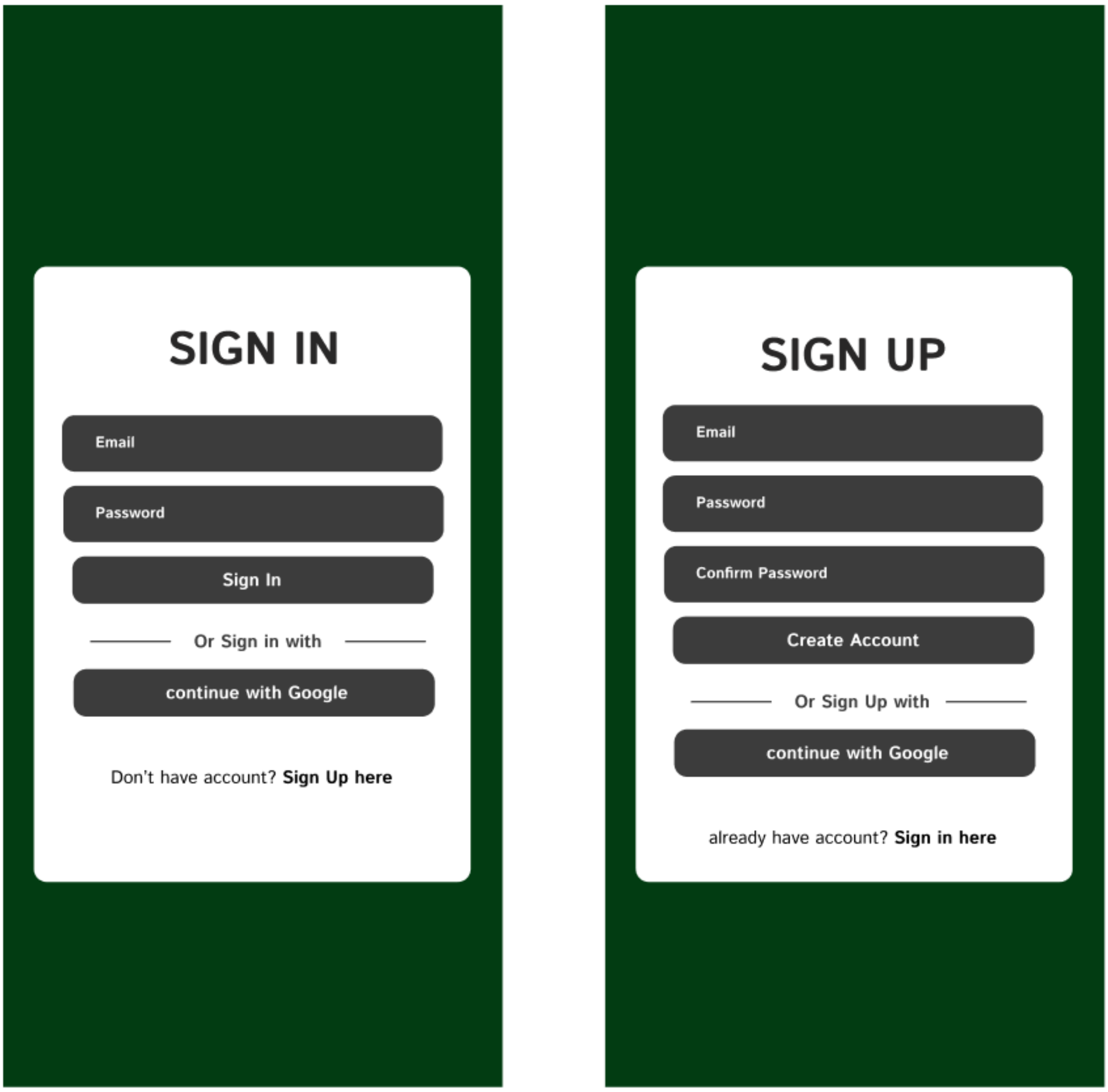
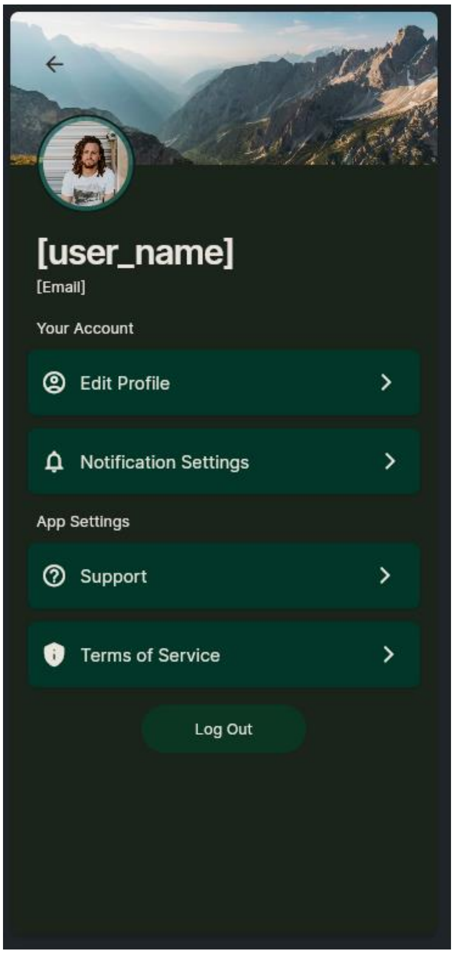


Figura 2 Seed Of Life’s Authentication Interfaces

Seed Of Life's Home Interface :

Figura 2 Seed Of Life's Homepage Interfaces

Seed Of Life's User Profile module interfaces :



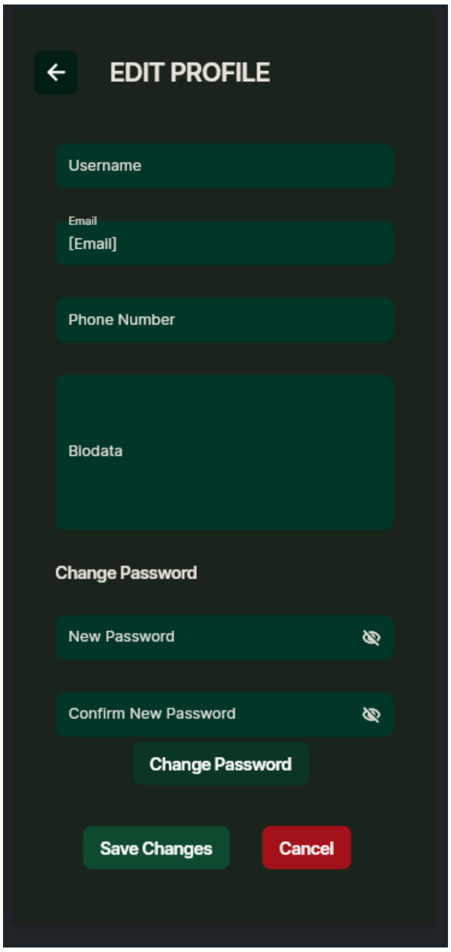


Figura 4 Seed Of Life's user profile module interfaces

Seed Of Life’s Plant Identification module interfaces:

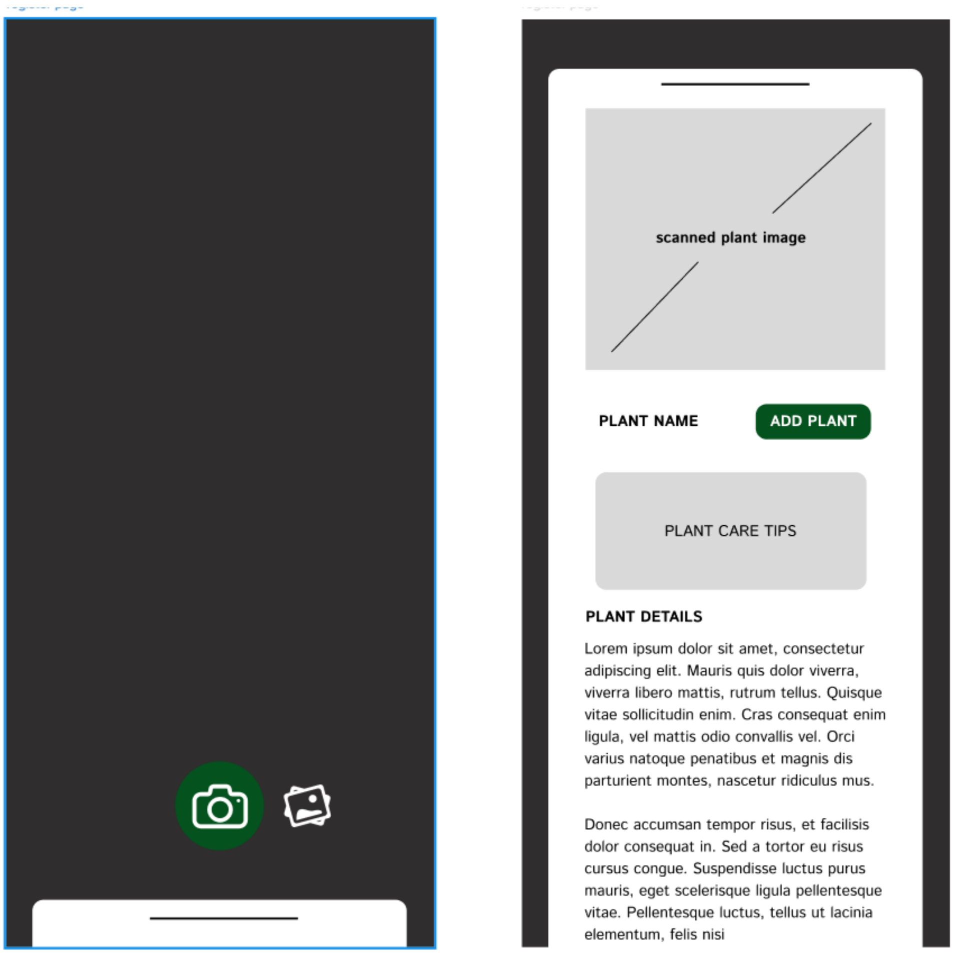


Figura 5 Seed Of Life’s PlantIdentification module interfaces

Seed of Life's Personalise Plant Care interfaces:

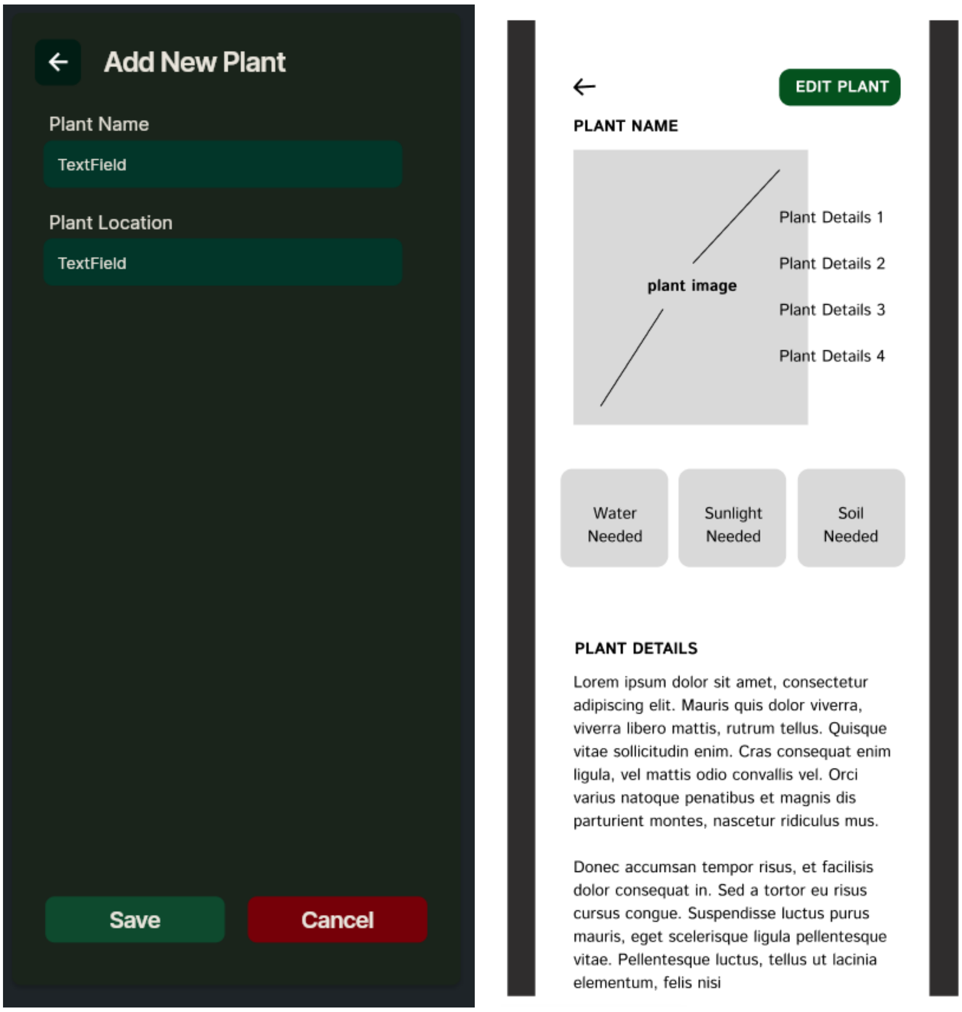


Figura 6 Seed Of Life's Personalise Plant Care module interfaces

Seed Of Life’s Community Page interfaces:

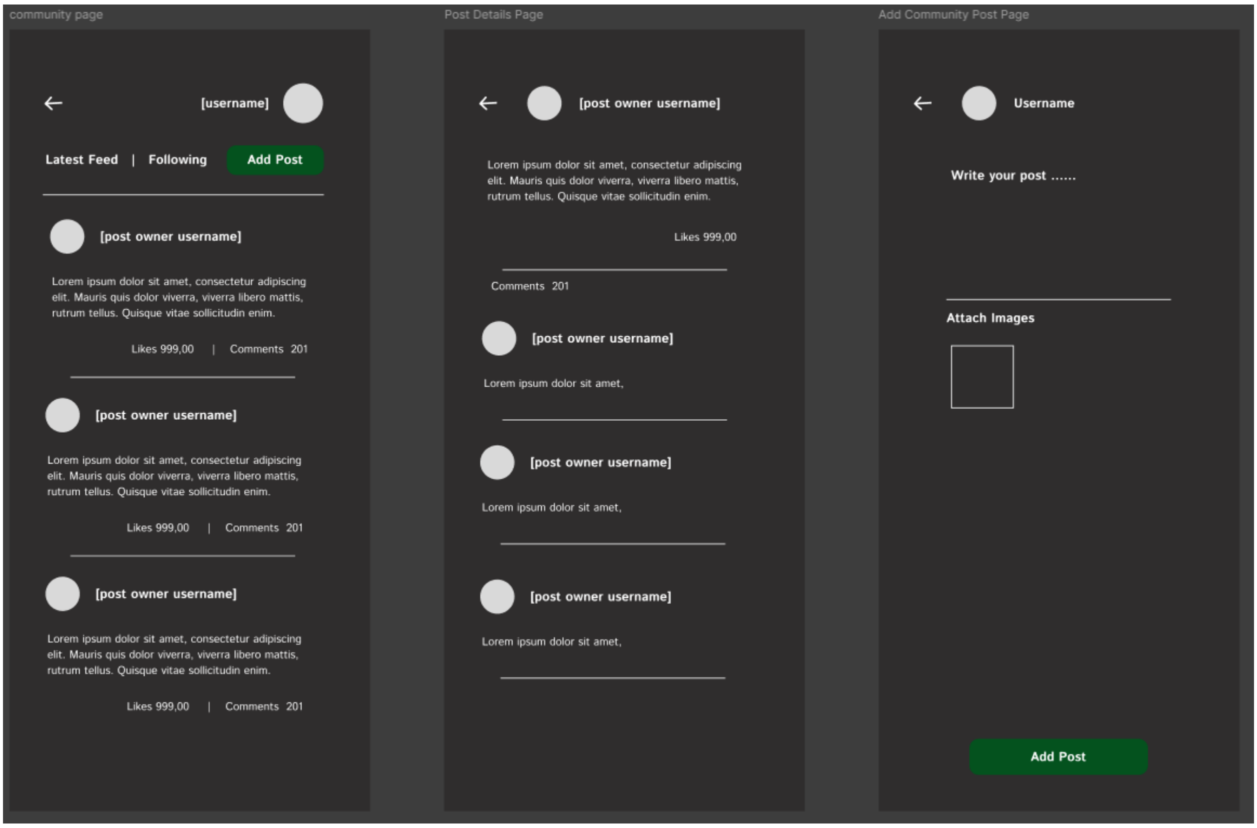


Figura 7 Seed Of Life’s Community module interfaces

Seed Of Life's Nursery Locator interface:

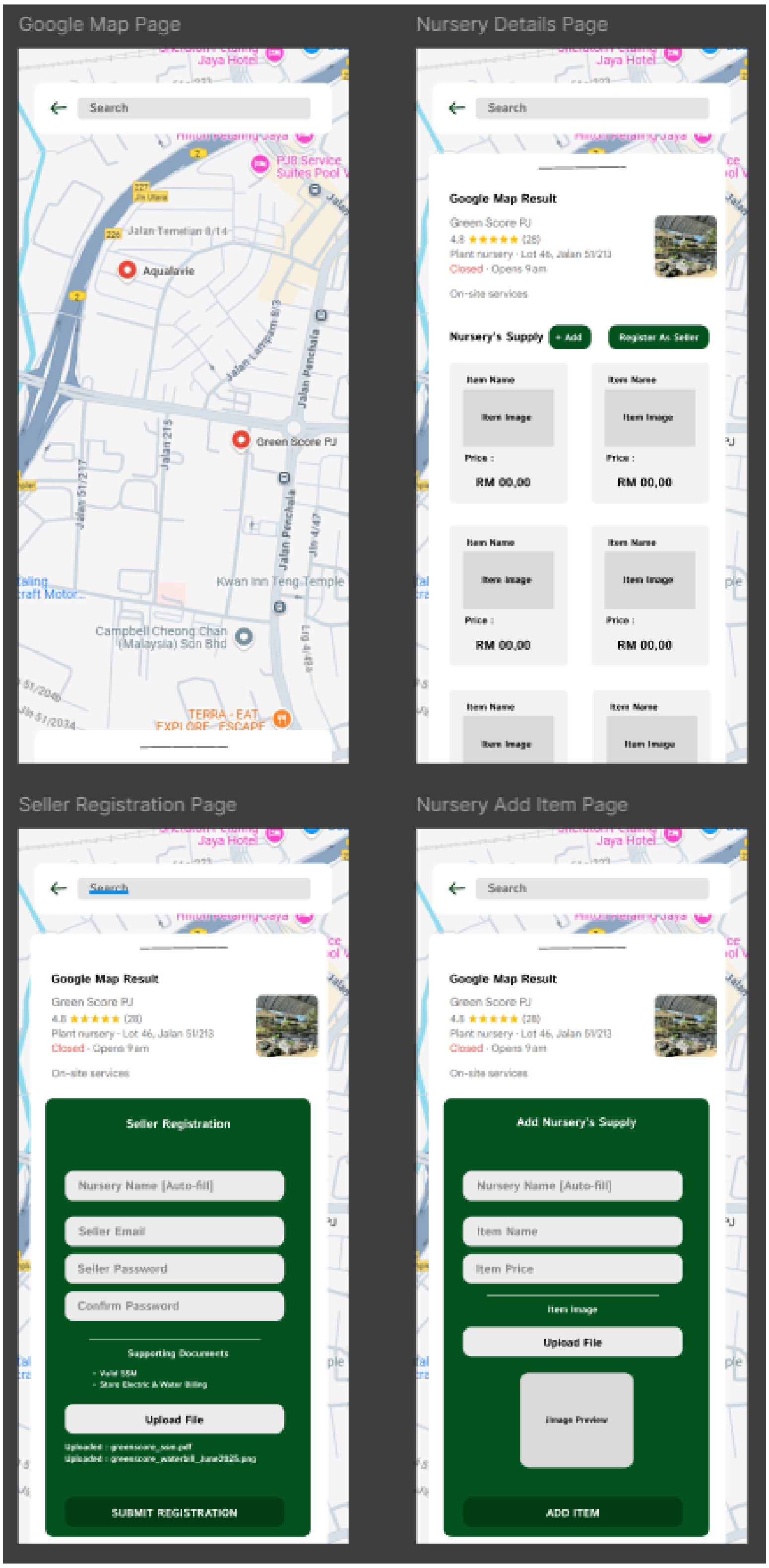


Figura 8 Seed Of Life's nursery locator module interfaces

4.2 Hardware Interfaces

Specifies hardware dependencies and interactions:

- Camera: For capturing plant photos (min. 8MP recommended).
- GPS: For nursery locator functionality.
- Storage: 100MB+ space for cached data.

4.3 Software Interfaces

Lists third-party software integrations critical for functionality:

- Plant Identification APIs: PlantNet, Perenual, Trefle.io and Plant.id for species recognition.
- Google Maps API: Nursery/shop location services.
- WeatherAPI: Real-time climate data for care adjustments.
- Firebase: Backend services (authentication, database).

4.4 Communications Interfaces

Defines protocols for data exchange:

- Network: Requires stable internet (3G/4G/Wi-Fi) for real-time features (e.g., API calls).
- Offline Mode: Cached data syncs when reconnected.
- Security: TLS 1.2+ encryption for data in transit (Section 5.3).

5. Other Non-functional Requirements

5.1 Performance Requirements

To ensure optimal user experience, SeedOfLife shall meet the following performance benchmarks:

A. Response Time

Feature	Requirement
Plant Identification	≤ 5 seconds for image processing under stable internet (3G/4G/Wi-Fi).
Database Queries (Plant Info)	≤ 2 seconds for loading pre-stored plant care details.
Forum Post Loading	≤ 3 seconds to display paginated community posts (20 posts per page).
Nursery Locator	≤ 4 seconds to load nearby stores on map (with GPS enabled).

Table 26 Response Time

B. Throughput & Scalability

Metric	Requirement
Concurrent Users	Support 1,000+ active users without significant latency (<10% slowdown).
API Requests	Handle 50+ requests/second during peak usage (e.g., monsoon season alerts).
Data Sync	Community posts/comments sync across devices within 10 seconds.

Table 26 Throughput & Scalability

C. Resource Usage

Component	Requirement
Memory Usage	≤ 150 MB RAM for mid-range Android devices (4GB RAM).
Battery Consumption	Background processes (e.g., notifications) shall not drain > 5% battery/hour.
Data Storage	≤ 50 MB for cached plant data and user profiles (expandable via cloud).

Table 27 Resource Usage

D. Offline Performance

Feature	Requirement
Cached Plant Details	Accessible without internet for recently viewed plants (72-hour expiry).
Care Reminders	Scheduled notifications fire offline, syncing when connection resumes.
Draft Forum Posts	Save locally and auto-submit when online.

Table 28 Offline Performance

E. Failover & Recovery

Scenario	Requirement
API Downtime	Fallback to cached plant database with warning banner.
App Crash	Auto-recover user session with ≤ 15-second reload time.
Data Loss Prevention	Daily encrypted backups of user profiles/plant collections.

Table 28 Failover & Recovery

Compliance Metrics:

- Performance tested on Android 10+ devices with 2GB+ RAM
- Benchmarks validated under 85% network reliability (typical Malaysian urban coverage).

5.2 Safety Requirements

To ensure user safety and system reliability, SeedOfLife shall adhere to the following safety requirements:

A. User Safety

Requirement	Description
Non-Toxic Plant Identification	The app shall include warnings for plants that are toxic to humans or pets (e.g., "This plant is harmful if ingested").
Safe Handling Guidance	For plants with thorns, irritants, or allergens, the app shall provide handling precautions (e.g., "Wear gloves when pruning").
Emergency Contacts	If a user searches for toxic plant care, the app shall display local poison control hotlines (Malaysia: +603-2694 3333).

Table 29 User Safety

B. Data & Privacy Safety

Requirement	Description
Secure Authentication	User passwords shall be encrypted (AES-256) and stored securely.
Location Privacy	GPS data for nursery locator shall not be stored beyond the active session unless explicitly permitted by the user.
Content Moderation	Community posts shall be scanned for harmful content (e.g., misinformation, self-harm) using AI + manual moderation.

Table 30 Data & Privacy Safety

C. System Safety

Requirement	Description
Misidentification Safeguards	If the AI cannot confidently identify a plant (confidence <80%), it shall display: "Verify with a local expert before handling."
Weather Alerts	During extreme weather (monsoons, heatwaves), the app shall adjust care reminders (e.g., "Delay watering due to heavy rain").

Offline Mode Limits	Critical features requiring real-time data (e.g., weather-based alerts) shall disable in offline mode with a clear warning.

Table 31 System Safety

D. Compliance & Legal

Requirement	Description
GDPR/Personal Data Protection Act (PDPA)	User data collection shall comply with Malaysia’s PDPA 2010. Users shall opt-in for analytics tracking.
Disclaimers	The app shall include: "Plant identification is AI-assisted; cross-check with experts for medical/edible plants."

Table 32 System Safety

5.3 Security Requirements

5.3.1 Data Protection

Encryption Standards

- All sensitive user data (passwords, personal information) shall be encrypted using AES-256
- Data in transit shall use TLS 1.2+ protocols

Authentication

- Implement multi-factor authentication for admin accounts
- Session tokens shall expire after 30 minutes of inactivity
- Password requirements:
  - Minimum 8 characters
  - At least 1 uppercase letter
  - At least 1 number or special character

Data Access Control

- Role-based access:
  - Regular users: Read/write own data only
  - Moderators: Additional content management privileges
  - Admins: Full system access
- API endpoints shall validate user permissions for each request

5.3.2 System Security

API Security

- Rate limiting of 100 requests/minute per IP address
- API keys required for PlantNet/Plant.id integrations
- Input validation for all API parameters

Vulnerability Management

- Regular security scans using OWASP ZAP
- Patch critical vulnerabilities within 72 hours of discovery
- Annual third-party penetration testing

Audit Logging

- Log all:
    - Authentication attempts
    - Data modification actions
    - Admin activities
- Retain logs for 90 days minimum

5.3.3 Privacy Compliance

Data Collection

- Explicit user consent required for:
    - Location data
    - Camera access
    - Push notifications
- Provide clear opt-out options

Data Retention

- User account data: Retained until account deletio
- Guest session data: Purged after 30 days
- Forum posts: Anonymized upon account deletion

Breach Notification

- Notify affected users within 72 hours of breach detection
- Comply with Malaysia's PDPA reporting requirements

D. Secure Development

Coding Practices

- Follow OWASP Top 10 guideline
- Static code analysis for each release
- Dependency scanning for known vulnerabilities

Environment Security

- Production databases isolated from development environments
- Secrets management using AWS Secrets Manager or equivalent
- Infrastructure as code with security groups limiting access

Verification Methods

- Automated security testing in CI/CD pipeline
- Quarterly security audits
- Bug bounty program for ethical hackers

5.4 Software Quality Attributes

5.4.1 Reliability

System Availability

- 99.5% uptime target for core services during business hours (8AM-12AM MYT)
- Graceful degradation for non-critical features during peak loads

Error Handling

- Automatic retry mechanism for failed API calls (max 3 attempts
- User-friendly error messages without technical details
- Critical errors logged with full context for debugging

Data Integrity

- Transaction rollback capability for database operation
- Checksum verification for plant database updates
- Daily integrity checks on user data stores

5.4.2 Maintainability

Code Quality

- Adherence to Flutter style guide
- Minimum 70% unit test coverage for critical modules
- Documentation for all public APIs and components

Modularity

- Clear separation between:
  - UI components
  - Business logic
  - Data access layers

- Feature flags for experimental functionality

Deployment

- Blue-green deployment capability
- Rollback procedure tested monthly
- Environment parity between staging and production

5.4.3 Usability

Accessibility

- WCAG 2.1 AA compliance
- Dynamic text sizing (110-150% without layout breakage)
- Sufficient color contrast (4.5:1 minimum)

Localization

- Support for Malay and English
- Culturally appropriate plant naming conventions
- Right-to-left text support for future Arabic localization

Learnability

- Interactive onboarding tutorial
- Contextual help tooltips
- Progressive disclosure of advanced features

5.4.4 Portability

Device Compatibility

- Support for Android 10+ (API level 29+)
- Responsive layout from 4.7" to 10" screens
- Adaptive icons and splash screens

Architecture

- Clean separation of platform-specific cod
- Dependency injection for service components
- Abstracted hardware interfaces (camera, GPS)

5.4.5 Efficiency

Resource Usage

- Cold start time <1.5 seconds
- Memory footprint <200MB on average
- Background sync interval configurable (15-60 minutes)

Network Optimization

- Compressed API payloads (GZIP)
- Smart prefetching of likely-needed data
- Delta updates for plant database

Quality Metrics:

- Monthly user satisfaction surveys (target ≥4/5 rating
- Crash-free sessions >99.8%
- App Store rating maintained ≥4.3 stars

6. Other Requirements

Appendix A: Glossary

Term	Definition
AI Model	Machine learning algorithm used for plant identification
APK	Android application package file format
Balcony Microclimate	Unique growing conditions on high-rise balconies (wind, sun exposure)
Care Schedule	Automated timeline of watering/fertilizing tasks
Crowdsourcing	User-contributed nursery locations and plant tips
Firebase	Backend service for authentication and database
Geofencing	Virtual perimeter for location-based notifications
IoT Integration	Future compatibility with smart plant sensors
Monsoon Mode	Special care adjustments for rainy seasons
Plant.id	Third-party plant identification API
PlantNet	Open-source plant recognition API
PDPA	Malaysia's Personal Data Protection Act 2010
SDG 11	UN Sustainable Development Goal for cities

Table 33 Glossary

Appendix B: Analysis Models

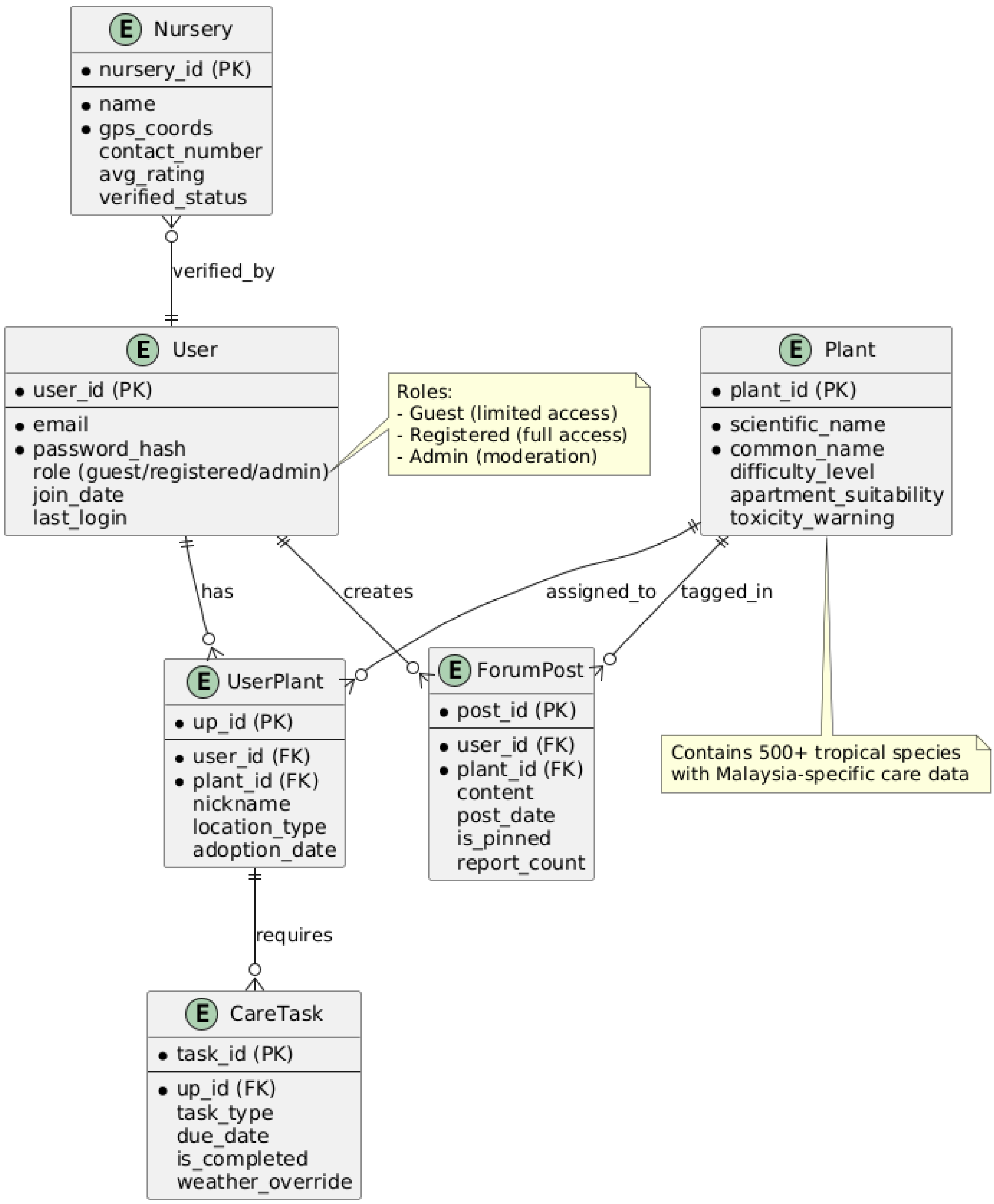


Figura 5 Analysis Models

Appendix C: Issues List

ID	Issue	Priority	Status	Owner
C-001	PlantNet API limited to 500 free calls/day	High	Pending	Dev Team
C-002	Malay language support incomplete	Medium	In Progress	Localization
C-003	GPS accuracy varies in dense urban areas	Medium	Researching	UX Team
C-004	Notification timing conflicts with prayer times	Low	Backlog	Product Owner
C-005	Data export feature requested by users	Medium	Approved	Backend
C-006	Rare tropical plants missing from database	High	Curating	Horticulturist

Table 34 Issues List